

When and Where Do Retail Promotions Work Best?

Evidence from Rossmann daily store sales

A transparent analysis of timing, store characteristics and competitor proximity.

Source observations	Stores	Historical coverage
1,017,209	1,115	2013 - July 2015

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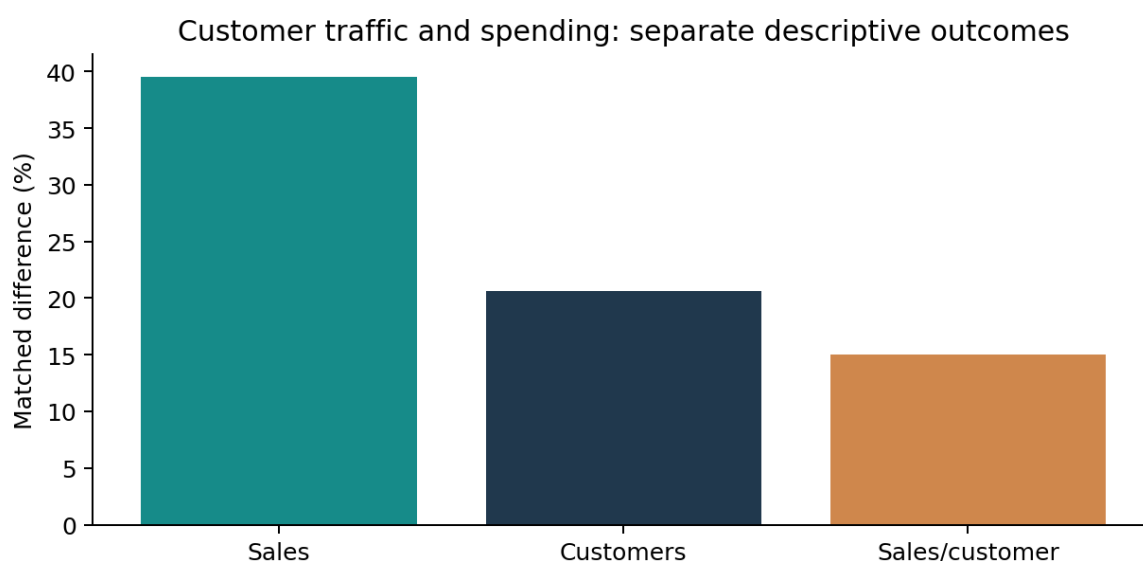
The findings describe promotion-associated differences. They do not establish causal uplift, net profit or an optimal promotion policy.

Executive summary

Within matched store-month-weekday comparisons, sales are 39.6% higher on promotion days. The corresponding differences are 20.6% for customer count and 15.0% for daily sales per customer. Both traffic and customer spending therefore accompany the sales difference; this is not a causal mediation decomposition.

Timing is the clearest recurring pattern. The matched sales difference is 64.3% on Monday and 21.3% on Friday. Monday remains the largest and Friday the smallest in each calendar year and in both held-out model scenario comparisons.

The nearest-competitor comparison is weaker. Stores within 500 metres show a somewhat larger matched difference than distant stores, but distance groups differ in format and the model specifications do not preserve a single proximity ranking. Distance should inform a test design rather than dictate a campaign.



Business implication

Prioritize a controlled test of weekday timing. Stratify the pilot by store format and competitor proximity, then evaluate incremental gross margin, offer costs and post-promotion sales. This historical analysis provides hypotheses for that test.

The word "difference" is deliberate. Promotion status is the same across stores on each observed weekday date, so the available data cannot separate promotion from all date-specific demand shocks.

1. Data and scope

The two source files are a daily sales panel and a store metadata table. Each daily observation records one store on one date. The join is validated as many-to-one; there are no duplicate Store-Date pairs and all daily rows have store metadata.

Audit item	Result
Total daily observations	1,017,209
Stores	1,115
Coverage	2013-01-01 to 2015-07-31
Closed-store rows	172,817
Open days with zero sales	54
Analysis population	695,791 store-days
Matched calendar strata	134,564
Rows in eligible matched strata	582,122 (83.7% of analysis rows)

The analysis population contains open stores with positive sales and customers, Monday-Friday, excluding public holidays. School holidays remain. This common population supports sales, traffic and sales/customer comparisons. It also avoids unsupported weekend promotion scenarios.

Store metadata contain three missing competitor distances and 354 missing competitor opening dates. Missing competitor dates are not converted into a claimed opening. Promo2 fields are structurally absent for non-participating stores. Snapshot metadata cannot fully reconstruct historical local competition.

The provided test.csv has no observed Sales and Customers outcomes. It and sample_submission.csv are retained in the private data bundle but are not used to answer the case-study questions.

2. A comparison that can be explained

Match within store and calendar context

Within each Store x YearMonth x DayOfWeek group, compare mean outcomes on promotion and non-promotion dates. Require at least two days of each status. This holds store identity, broad seasonality and weekday fixed, while retaining multiple dates per comparison.

Percentage difference = $100 \times (\text{mean outcome on promo days} / \text{mean outcome on non-promo days} - 1)$. Average over eligible strata within each store, then average equally over stores. This estimates a typical-store descriptive contrast rather than a chain revenue-weighted return.

Three outcomes

Outcome	Interpretation
Sales	Daily turnover, in the dataset's sales units.
Customers	Reported customer count for the day.
Sales/customer	Mean daily spending per reported customer, not transaction-level basket size.

The three percentage contrasts are separately averaged. They should not be added or multiplied as an exact decomposition of the overall sales contrast.

What the intervals cover

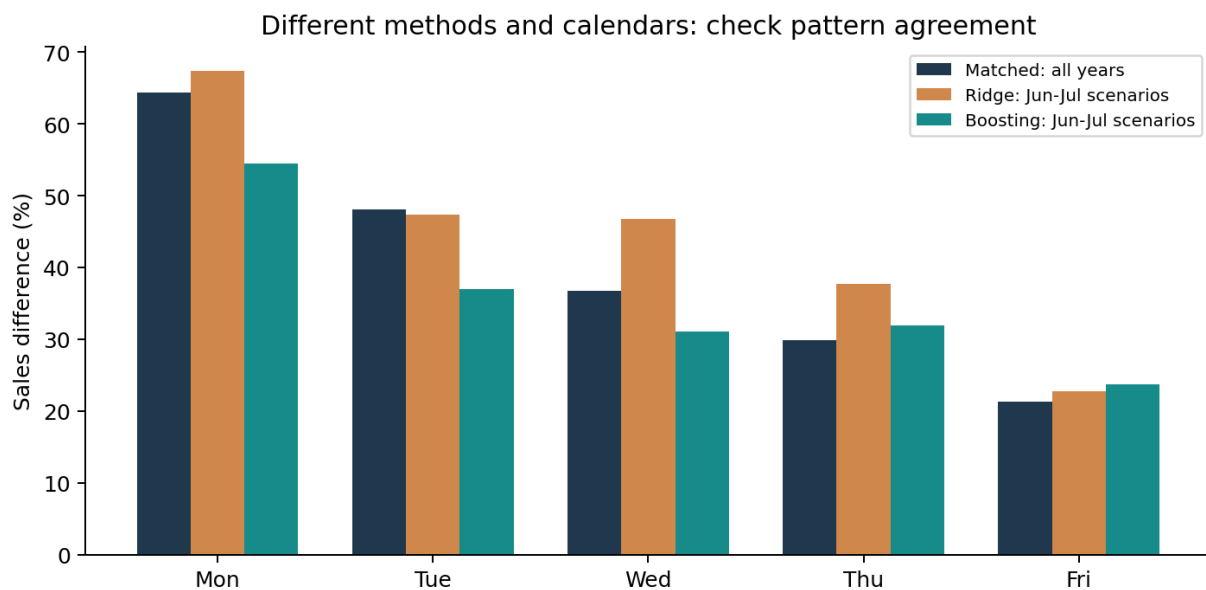
Resample whole stores 500 times and recompute the mean store-level contrast. The plotted 95% intervals describe cross-store variation conditional on this calendar. They do not include common date shocks, future-calendar uncertainty or unmeasured confounding. No causal significance claim is made.

What remains uncontrolled

Exact-date demand, promotion choice, offer depth, local events and lagged demand may affect outcomes. Promotions are synchronized across stores on all 674 weekday dates. Exact-date fixed effects would absorb the promotion indicator, making a separate promotion coefficient unidentified in that design.

3. Weekday timing is the strongest pattern

Weekday	Matched sales difference	Model scenario: boosting
Monday	64.3%	54.4%
Tuesday	48.1%	37.0%
Wednesday	36.8%	31.1%
Thursday	29.9%	32.0%
Friday	21.3%	23.7%



Monday is consistently the largest difference and Friday the smallest. The exact middle-week order is less stable: Thursday slightly exceeds Wednesday in the boosting scenarios for June-July. That disagreement is useful evidence against an overly precise daily ranking.

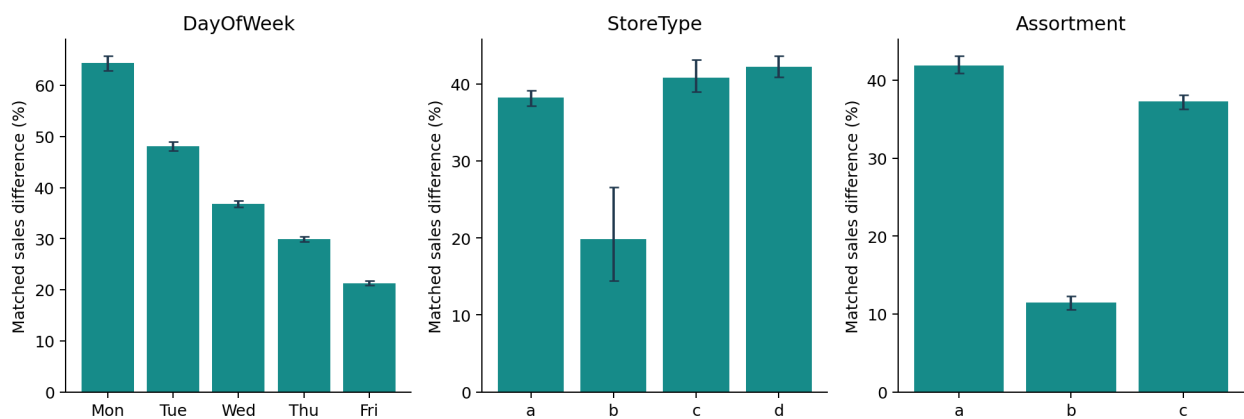
The matched estimates pool 2013-July 2015; model scenarios use June-July 2015 only. Their magnitudes are not directly interchangeable. Agreement in the broad timing pattern is more informative than agreement to a decimal place.

Managerial reading: compare early-week and late-week campaigns in a controlled pilot. The dataset supports a timing hypothesis; it does not prove that moving a promotion to Monday increases annual profit.

4. Store format and assortment

Store type	Stores	Matched sales difference
a	602	38.2%
b	17	19.8%
c	148	40.8%
d	348	42.3%

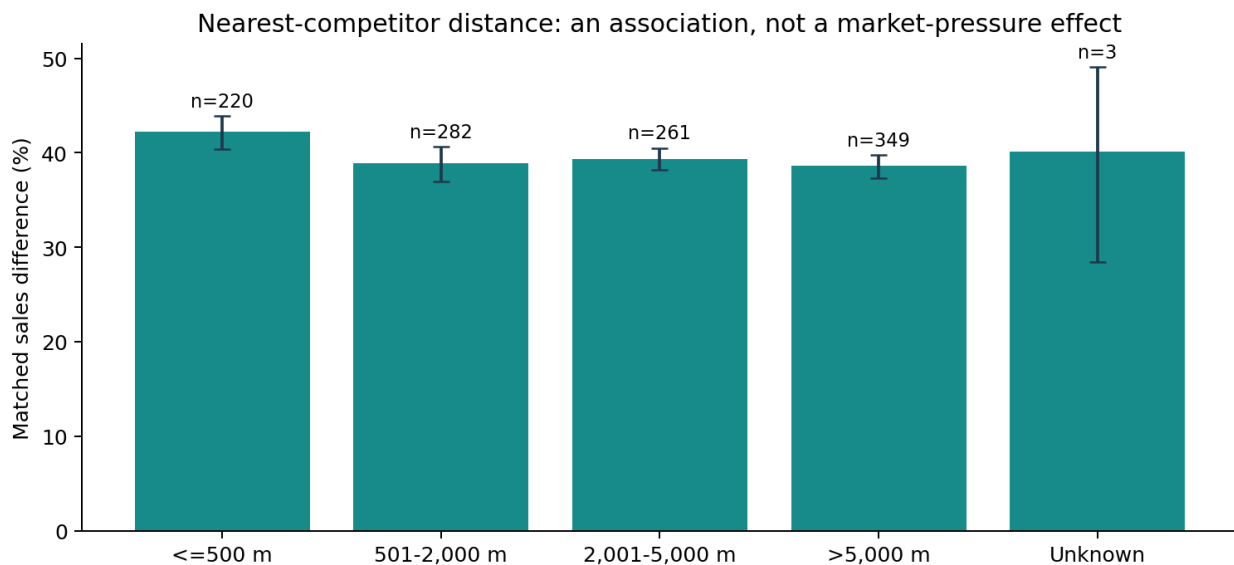
Assortment code	Stores	Matched sales difference
a	593	41.9%
b	9	11.4%
c	513	37.3%



Types c and d show somewhat larger matched differences than type a. Type b is much lower, but contains only 17 stores. Assortment b contains only nine stores. Those small groups are insufficient for broad operational recommendations, even when store-resampling intervals look narrow.

The store formats are anonymized. Assortment a is basic, b extra and c extended according to the data dictionary. The analysis does not imply that changing a store's format or expanding assortment would cause a change in promotion response.

5. Competitor proximity: a modest association



The matched sales difference is about 42.2% at distances up to 500 metres and 38.6% beyond 5 kilometres. The intermediate groups are around 39%. The nearest-versus-farthest gap is roughly 3.6 percentage points, much smaller than the weekday contrast.

Restricting to stores whose recorded competitor opening predates 2013 leaves the nearest-versus-farthest pattern modest: about 41.9% versus 39.1%. This restriction addresses some timing ambiguity but cannot reconstruct past competitor locations.

Why the strategic interpretation stays cautious

Formats and locations differ across distance groups. Within-format tables are included in results/competition_within_format.csv so the pooled relationship can be inspected. They should not be interpreted as a randomized comparison of competition.

In June-July scenarios, boosting produces about 37.1% for the nearest group and 35.5% for the farthest. Ridge gives about 42.2% and 43.9%, reversing those endpoints. Since Ridge has no explicit proximity-by-promotion interaction, its percentage contrasts partly reflect different predicted baseline sales.

Managerial reading: include proximity as a stratification variable in a pilot, but do not increase discounting solely because the nearest competitor is close. CompetitionDistance is a distance measure, not a complete measure of competitive pressure.

6. Predictive models and temporal evaluation

Ridge regression is the readable baseline. It includes one-hot store identifiers and calendar categories, standardized numeric predictors, and promotion-by-weekday interactions. An L2 penalty of 100 controls coefficient size. Unseen year-month categories in the test period are handled as unknown, which is a limitation of this baseline.

Histogram gradient boosting is the nonlinear benchmark. It uses 2013 store-average sales as a fixed historical predictor rather than a numeric store ID. It contains no current-day customer outcome. Native categorical handling is used for store type, assortment, month and weekday.

Period	Role	Rows
2013	Fixed store-history baseline	Not model-training outcomes
Jan 2014-Mar 2015	Fit candidate models	325,718
Apr-May 2015	Select tree complexity	42,353
Jun-Jul 2015	Final holdout	49,355

Two tree sizes, 15 and 31 leaves, were evaluated on validation MAE. The 31-leaf model was selected. Both models were refit through May before the final test. Boosting uses 180 iterations, learning rate 0.08, minimum 100 observations per leaf, L2 regularization 10 and random seed 42. Internal random early stopping is disabled.

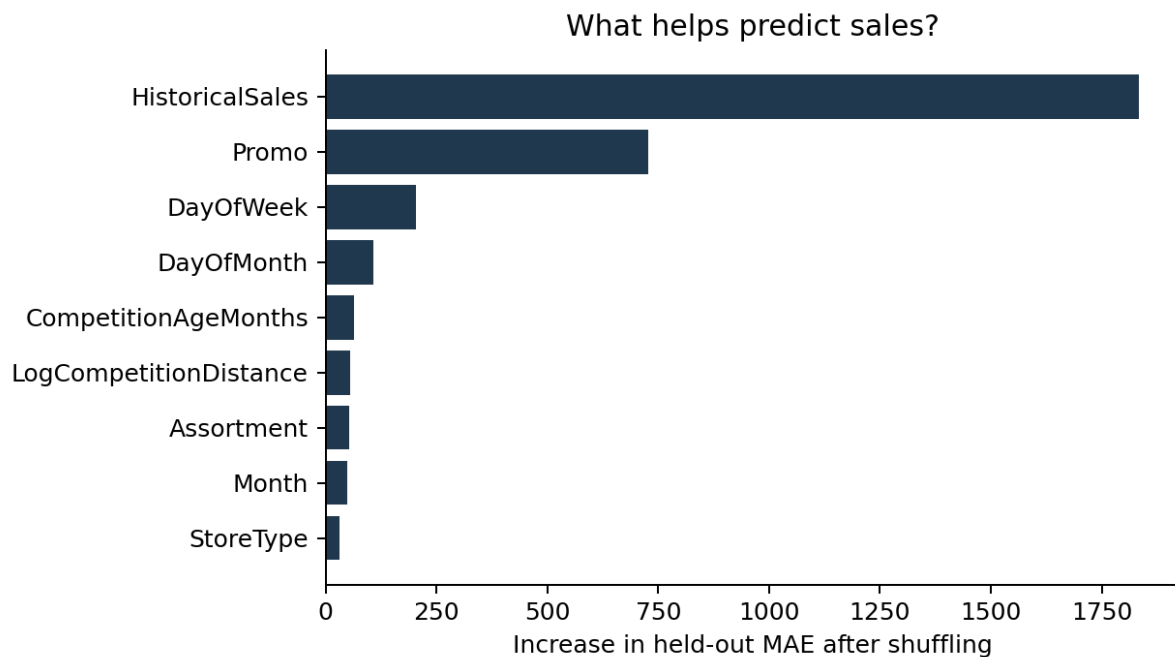
Final test model	MAE	RMSE	R-squared
Ridge	859.0	1,196.2	0.850
Histogram boosting	797.7	1,095.3	0.874

Boosting reduces test MAE by 7.1% relative to Ridge. These errors are in sales units; currency is not asserted. Sales forecast accuracy supports the benchmark, but it does not validate an unobserved promotion counterfactual.

7. Scenario contrasts and model explanation

For each eligible held-out store-day, the fitted model predicts sales twice: once with Promo=0 and once with Promo=1, keeping other pre-outcome features fixed. The relative difference is averaged within stores and then across stores. Scenario rows are exported for inspection.

The contrast is model-implied, not observed causal uplift. It assumes other included conditions stay fixed and says nothing about changing discount depth, competitors reacting or subsequent sales. No Promote/Skip threshold is imposed in this first project.



Permutation importance is measured on a reproducible sample of 6,000 held-out observations, with three shuffles per predictor. It records how much MAE increases when that predictor is shuffled. The fixed historical store mean and promotion status are useful predictive information; correlated variables can split or mask importance.

Importance ranks predictors for forecasting. It does not estimate the causal importance of promotion, nor does it demonstrate an interaction between promotion and competition. The matching tables and scenario comparisons carry that discussion separately.

8. Robustness and practical conclusions

Check	Stores	Matched sales difference
2013	1115	37.8%
2014	1115	42.1%
2015 Jan-Jul	1115	38.8%
At least 5 total days	1115	44.7%
Exclude top 1% sales	1114	39.2%

Year-specific magnitudes vary, but the positive aggregate association remains. Excluding the largest 1% of observed sales changes the main contrast only modestly. Requiring five total dates per stratum raises the contrast to 44.7%, about 5.1 percentage points above the main estimate. That restriction changes calendar composition, so the magnitude is sensitive to support requirements even though the overall association stays positive. The trimming exercise drops one store with no remaining eligible pairs.

Leaving each calendar month out gives overall contrasts from 38.9% to 40.9%. Monday remains highest and Friday lowest in each separately summarized year; middle-week rankings vary.

Five implications that follow from the evidence

1. Test weekday timing first. It produces a clearer and more stable contrast than competitor distance.
2. Monitor traffic and sales per customer together. Both change with promotion, so a sales-only dashboard misses useful context.
3. Stratify by store format. Treat type b and assortment b as small-sample cases requiring local review.
4. Use competitor distance as a hypothesis, not a discounting rule. The proximity pattern is small and sensitive to model specification.
5. Evaluate margin and demand after the campaign. A sales gain may be offset by discount costs or purchases shifted from nearby days.

9. Limits, reproduction and next test

Limits of the evidence

There is no random assignment, and chain-wide promotion timing prevents identification with exact-date fixed effects. Store-resampling intervals exclude common calendar shocks. Positive-outcome filtering defines the target population; closed days and weekend promotions are outside the claim. Competition metadata may be a snapshot. Small segments and the 2014 store-coverage interruption can affect calendar composition.

No gross margin, promotion cost, discount depth, product mix, competitor prices or customer identities are available. The data cannot identify long-term promotion dependence, fatigue, new-customer acquisition, transaction basket size or profit-maximizing conditions. Customers can include repeat visitors. All findings concern the historical panel, not current retail conditions.

Reproduction

Run notebooks/promotion_strategy.ipynb in cell order or execute `python src/run_analysis.py` from the project root. Provide `train.csv` and `store.csv` in `data/` or set `ROSSMANN_DATA_DIR`. `requirements.txt` records analysis package versions; `results/input_manifest.json` records file hashes. CSV tables, prediction rows and plots are exported automatically.

The Kaggle notebook contains the same analysis and executed results, with automatic attached-data discovery. The public-ready archive omits raw data. The private archive includes the supplied files; competition terms still govern any redistribution.

Next test

Run a randomized store-week pilot comparing early-week and late-week promotion schedules, stratified by format and proximity. Define gross margin net of offer cost as the primary outcome, include a post-promotion window, and account for clustering and possible spillovers. This is a proposed experiment, not a result from this dataset.

Sources

Rossmann Store Sales, Kaggle (2015). Data dictionary and competition:
<https://www.kaggle.com/competitions/rossmann-store-sales/data>

scikit-learn documentation. HistGradientBoostingRegressor and model evaluation:
<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.HistGradientBoostingRegressor.html>

All numerical findings were calculated from the user-supplied `train.csv` and `store.csv`. No simulated values were substituted for observations.